Reading a Rational Graph

Simplifying and finding the domain, intercepts, holes and vertical asymptotes, and horizontal asymptotes from the degrees

Directions

- Problems 1–4 warm up one skill at a time. From problem 5 on they are mixed, so decide for yourself which tool each one needs.
- Always factor the numerator and the denominator first. A cancelled factor gives a **hole**; a surviving denominator factor gives a **vertical asymptote**.
- An x-intercept is a zero of the *simplified* function; the y-intercept is the value at $x = 0$.
- A horizontal asymptote is decided by the degrees: equal degrees give the ratio of the leading coefficients.

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1. Write $\dfrac{x^2 - 4}{x^2 + 5x + 6}$ in fully simplified form.

Answer: ________________

2. Let $f(x) = \dfrac{x - 4}{x + 1}$.

(a) Find the x-intercept: ________ **(b)** Find the y-intercept: ________

3. Let $g(x) = \dfrac{x^2 - x - 6}{x^2 - 9}$.

(a) Write $g(x)$ in fully simplified form: _________

(b) Give the equation of the vertical asymptote: _________

4. Find the equation of the horizontal asymptote of $h(x) = \dfrac{3x^2 - x}{x^2 + 4}$.

Answer: __________________

5. Let $u(x) = \dfrac{x^2 - 9}{x^2 - x - 12}$.

(a) Find every x-intercept of the graph of u: _________

(b) Find the y-intercept: _________

6. Let $v(x) = \dfrac{x^3 - 9x}{x^2 + 2x - 15}$.

(a) Write $v(x)$ in fully simplified form: _________

(b) List the values of x excluded from the domain of v: _________

7. Let $w(x) = \dfrac{x^2 - 5x + 6}{x^2 - 4}$.

(a) The graph has one hole. Find its height: _________

(b) Give the equation of the vertical asymptote: _________

8. Find the equation of the horizontal asymptote of $r(x) = \dfrac{2x + 7}{5x - 1}$.

Answer: __________________

9. Let $s(x) = \dfrac{2x^2 - 8}{x^2 + 1}$.

(a) Find every x-intercept: _________

(b) Find the y-intercept: _________

10. Let $t(x) = \dfrac{x^2 - 16}{x^3 - 4x^2}$.

(a) Write $t(x)$ in fully simplified form: _________

(b) List the values of x excluded from the domain of t: _________

11. Let $k(x) = \dfrac{4x^2 + 1}{2x^2 - 3x}$.

(a) Give the equation of the horizontal asymptote: ________

(b) Give the equations of the vertical asymptotes: ________

(c) Explain what the degrees of the numerator and denominator tell you about the horizontal asymptote, and where its height comes from.

12. **Synthesis.** Let $F(x) = \dfrac{x^3 - x^2 - 6x}{x^2 - 9}$.

(a) Write $F(x)$ in fully simplified form: _________

(b) Find every x-intercept: _________

(c) The graph has one hole. Find its height: _________

(d) Give the equation of the vertical asymptote: _________

(e) Explain why $x = 3$ does *not* give a vertical asymptote, even though it makes the original denominator zero.